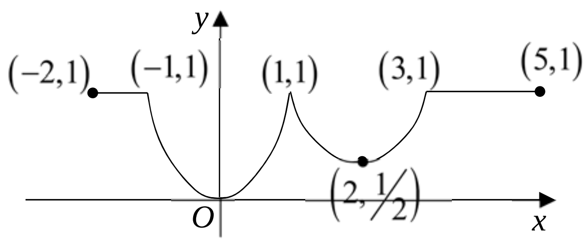
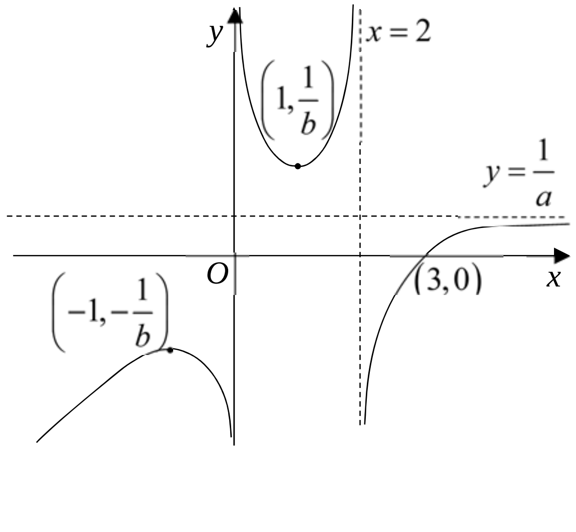
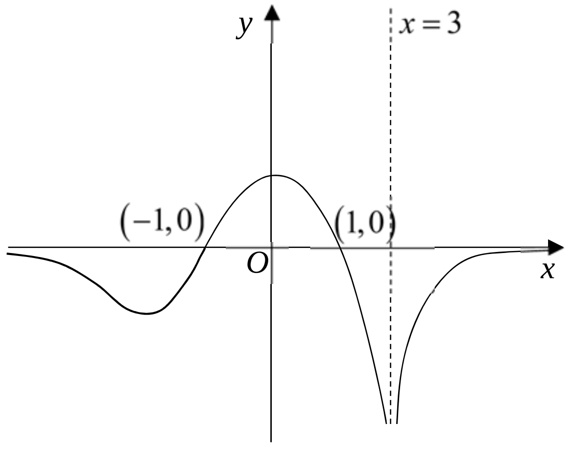
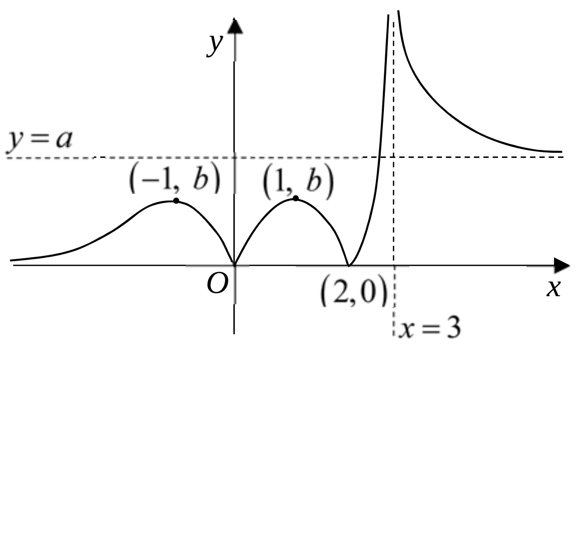
Yishun Junior College
2018 JC2 H2 Math Preliminary Examination
Paper 1 Solutions

Qns	Solutions
1	The coefficients of the polynomial (quadratic) equation are not real. Hence the Conjugate Root Theorem cannot be applied. $w^2-(6-5\mathrm{i})w+11-\mathrm{i}=0$ $w=\dfrac{(6-5\mathrm{i})\pm\sqrt{(6-5\mathrm{i})^2-4(1)(11-\mathrm{i})}}{2}$ $=\dfrac{(6-5\mathrm{i})\pm(4-7\mathrm{i})}{2}$ $=5-6\mathrm{i}$ or $1+\mathrm{i}$ Hence the other root is $5-6\mathrm{i}$.
2	(i) 1. A translation of a units in the direction of the x-axis. 2. A scaling parallel to the y-axis by a factor of $\frac{1}{a}$. 3. A translation of $\frac{1}{a}$ units in the direction of the y-axis. (ii) 

3	(i) $z-3=-2+\mathrm{i}a$ $\arg(z-3)=\pi-\tan^{-1}\left(\frac{a}{2}\right)$ $\pi-\tan^{-1}\left(\frac{a}{2}\right)=\frac{3\pi}{4}$ $\tan^{-1}\left(\frac{a}{2}\right)=\frac{\pi}{4}$ $\frac{a}{2}=1$ $a=2$ (ii) $\left\lvert\frac{w^n}{w^*}\right\rvert=\frac{\lvert w\rvert^n}{\lvert w^*\rvert}=\frac{\lvert w\rvert^n}{\lvert w\rvert}=\frac{b^n}{b}=b^{n-1}$ (iii) $\arg\left(\frac{w^n}{w^*}\right)=\arg\left(w^n\right)-\arg\left(w^*\right)$ $=n\arg(w)-(-\arg(w))$ $=n\arg(w)+\arg(w)$ $=n\left(\frac{\pi}{2}\right)+\left(\frac{\pi}{2}\right)$ $=\left(\frac{\pi}{2}\right)(n+1)$ Since $\frac{w^n}{w^*}$ is purely imaginary. $\arg\left(\frac{w^n}{w^*}\right)=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2},\ldots$ $\left(\frac{\pi}{2}\right)(n+1)=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2},\ldots$ $n+1=1,3,5,\ldots$ $\therefore$ two smallest positive whole number $n=2,4$

4	(a) $y=\dfrac{1}{\mathrm{f}(x)}$,  (b) $y=\mathrm{f}'(x)$  (c) $y=\lvert\mathrm{f}(x)\rvert$ 

5	(i) $y^3 = 7 + \cos x$ $3y^2 \dfrac{dy}{dx} = -\sin x$ $6y\left(\dfrac{dy}{dx}\right)^2 + 3y^2 \dfrac{d^2y}{dx^2} = -\cos x$ $6y\left(\dfrac{dy}{dx}\right)^2 + 3y^2 \dfrac{d^2y}{dx^2} = -\left(y^3 - 7\right)$ $6y\left(\dfrac{dy}{dx}\right)^2 + 3y^2 \dfrac{d^2y}{dx^2} = 7 - y^3$ (ii) When $x = 0$, $y = 2$, $\dfrac{dy}{dx} = 0$, $\dfrac{d^2y}{dx^2} = -\dfrac{1}{12}$ $6\left(\dfrac{dy}{dx}\right)^3 + 12y\left(\dfrac{dy}{dx}\right)\left(\dfrac{d^2y}{dx^2}\right) +$ $6y\left(\dfrac{dy}{dx}\right)\left(\dfrac{d^2y}{dx^2}\right) + 3y^2\left(\dfrac{d^3y}{dx^3}\right) = -3y^2 \dfrac{dy}{dx}$ When $x = 0$, $\dfrac{d^3y}{dx^3} = 0$ Since $\dfrac{d^3y}{dx^3} = 0$, term in $x^3 = \dfrac{f'''(0)}{3!}x^3 = 0$. Hence Maclaurin series of y does not have a term in x^3. (iii) $y = 2 + \left(-\dfrac{1}{12}\right)\dfrac{x^2}{2!} + \ldots$ $= 2 - \dfrac{1}{24}x^2 + \ldots$

	(iv) $y^3 = 7 + \cos x$ $y = \left[7 + \left(1 - \frac{x^2}{2} + \ldots\right)\right]^{\frac{1}{3}}$ $y = \left(8 - \frac{x^2}{2} + \ldots\right)^{\frac{1}{3}}$ $= 2\left(1 - \frac{x^2}{16} + \ldots\right)^{\frac{1}{3}}$ $= 2\left[1 + \frac{1}{3}\left(-\frac{x^2}{16}\right) + \ldots\right]$ $= 2 - \frac{1}{24}x^2 + \ldots$
6	(i) $\sum_{r=1}^{n} \ln\left(\frac{re^{r^2}}{r+1}\right) = \sum_{r=1}^{n}\left[\ln r + \ln e^{r^2} - \ln(r+1)\right]$ $= \sum_{r=1}^{n}\left[\left(\ln r - \ln(r+1)\right) + r^2\right]$ $= \sum_{r=1}^{n} \ln r - \ln(r+1) + \sum_{r=1}^{n} r^2$ $\begin{aligned} = & \quad \ln 1 \quad \cancel{-\ln 2} \\ & + \cancel{\ln 2} \quad \cancel{-\ln 3} + \frac{n}{6}(n+1)(2n+1) \\ & + \cancel{\ln 3} \quad \cancel{-\ln 4} \\ & \quad \vdots \\ & + \cancel{\ln(n-1)} \quad \cancel{-\ln n} \\ & + \cancel{\ln n} \quad -\ln(n+1) + \frac{n}{6}(n+1)(2n+1) \end{aligned}$ $= \frac{n}{6}(n+1)(2n+1) - \ln(n+1)$ (ii) $v_1 = 2^{\ln(u_1) - 1^2} = \frac{2^{\ln(u_1)}}{2^{1^2}}$ $v_2 = 2^{\ln(u_2) - 2^2} = \frac{2^{\ln(u_2)}}{2^{2^2}}$ …

$$v_n = 2^{\ln(u_n)-n^2} = \frac{2^{\ln(u_n)}}{2^{n^2}}$$

$$v_1 \times v_2 \times \cdots \times v_n = \frac{2^{\ln(u_1)} \times 2^{\ln(u_2)} \times \cdots \times 2^{\ln(u_n)}}{2^1 \times 2^4 \times \cdots \times 2^{n^2}}$$

$$= \frac{2^{\sum_{r=1}^{n} \ln(u_r)}}{2^{\sum_{r=1}^{n} r^2}}$$

$$= \frac{2^{\frac{n}{6}(n+1)(2n+1)-\ln(n+1)}}{2^{\frac{n}{6}(n+1)(2n+1)}}$$

$$= 2^{-\ln(n+1)}$$

$$v_1 \times v_2 \times \cdots \times v_n = \frac{1}{2^{\ln(n+1)}}$$

As $n \to \infty, \frac{1}{2^{\ln(n+1)}} \to 0$, therefore convergent.

7

(ai) $\int \cos 3x \cos x \ \mathrm{d}x = \frac{1}{2}\int 2\cos 3x \cos x \ \mathrm{d}x$

$$= \frac{1}{2}\int (\cos 4x + \cos 2x) \ \mathrm{d}x$$

$$= \frac{1}{2}\left(\frac{1}{4}\sin 4x + \frac{1}{2}\sin 2x\right) + c$$

$$= \frac{1}{8}\sin 4x + \frac{1}{4}\sin 2x + c$$

(aii) $\int_0^{\frac{\pi}{4}} x\cos 3x \cos x \ \mathrm{d}x = x\left[\frac{1}{8}\sin 4x + \frac{1}{4}\sin 2x\right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \left(\frac{1}{8}\sin 4x + \frac{1}{4}\sin 2x\right) \ \mathrm{d}x$

$$= \left(\frac{\pi}{4}\right)\left[\frac{1}{8}\sin 4\left(\frac{\pi}{4}\right) + \frac{1}{4}\sin 2\left(\frac{\pi}{4}\right)\right] - \left[-\frac{1}{32}\cos 4x - \frac{1}{8}\cos 2x\right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{\pi}{4}\right)\left[\frac{1}{8}\sin \pi + \frac{1}{4}\sin\left(\frac{\pi}{2}\right)\right] +$$

$$\left[\frac{1}{32}\cos 4\left(\frac{\pi}{4}\right) + \frac{1}{8}\cos 2\left(\frac{\pi}{4}\right) - \left(\frac{1}{32} + \frac{1}{8}\right)\right]$$

$$= \frac{\pi}{16} + \left[-\frac{1}{32} - \frac{1}{32} - \frac{1}{8}\right] = \frac{\pi - 3}{16}$$

	(b) $\int_{-a}^{a} \left\| x(x-2a) \right\| \, dx$ $= \int_{-a}^{0} x(x-2a)\ dx + \int_{0}^{a} -x(x-2a)\ dx$ $= \left[\frac{x^3}{3} - ax^2 \right]_{-a}^{0} - \left[\frac{x^3}{3} - ax^2 \right]_{0}^{a}$ $= 2a^3$
8	(i) $S_n \geq 10000$ $\frac{n}{2}[2(700)+(n-1)(500)] \geq 10000$ $\frac{n}{2}(900+500n) \geq 10000$ $500n^2 + 900n - 20000 \geq 0$ From GC, $n \leq -7.2883$ or $n \geq 5.4883$ (rejected) The train has completed 5 cycles $S_5 = \frac{5}{2}[2(700)+4(500)]$ $= 8500$ The train broke down after 1500km in the 6th cycle The train will run 600 km on the first day of the 6th cycle. The train will break down on the 3rd day of the 6th cycle. (ii) Let O_n be the sum of distance the train runs on *n* odd numbered days. $O_n = 100 + 0.92(1.2)(100) + 0.92^2(1.2)^2(100)$ $+ ... + 0.92^{n-1}(1.2)^{n-1}(100)$ $= \frac{100(1.104^n - 1)}{1.104 - 1}$ $= 961.54(1.104^n - 1)$ Let E_n be the sum of distance the train runs on *n* even numbered days $E_n = (1.2)100 + 0.92(1.2)^2(100) + 0.92^2(1.2)^3(100)$ $+ ... + 0.92^{n-1}(1.2)^n(100)$ $= \frac{120(1.104^n - 1)}{1.104 - 1}$ $= 1153.8(1.104^n - 1)$

Let S_n be the total distance covered by *n* days.

$$S_{2n} = O_n + E_n$$
$$= 961.54\left(1.104^n - 1\right) + 1153.8\left(1.104^n - 1\right)$$
$$= 2115.4\left(1.104^n - 1\right)$$

(iii)

$$S_{2n} > 10000$$
$$2115.4\left(1.104^n - 1\right) > 10000$$
$$1.104^n > 5.7275$$
$$n > 17.640$$

Therefore, the train must complete at least <u>36</u> days of travelling.

9

(i)

$$l_1: \quad x + 2 = \frac{z}{-2}$$

Let $\lambda = x + 2 = \dfrac{z}{-2}$

$$x = -2 + \lambda$$
$$y = 0$$
$$z = -2\lambda$$

$$\mathbf{r} = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$\left[\begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}\right] \cdot \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$$
$$= -8 + 4\lambda - 4\lambda$$

$= -8$ for all values of λ.

Hence all points on l_1 lies in p_1.

Hence, l_1 lies in p_1.

(ii)

Let the foot of perpendicular from *A* to p_1 be *N*.

Equation of the line that passes through *A* and perpendicular to p_1 is

$$l_{AN}: \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix} + \alpha \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}, \quad \alpha \in \mathbb{R}.$$

Since N lies on l_{AN}, $\overrightarrow{ON} = \begin{pmatrix} 4+4\alpha \\ 3-\alpha \\ k+2\alpha \end{pmatrix}$, for some $\alpha \in \mathbb{R}$

$$\begin{pmatrix} 4+4\alpha \\ 3-\alpha \\ k+2\alpha \end{pmatrix} \bullet \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = -8$$

$$16+16\alpha-3+\alpha+2k+4\alpha=-8$$

$$21\alpha=-21-2k$$

$$\alpha=-1-\frac{2k}{21}$$

$$\therefore \overrightarrow{ON} = \begin{pmatrix} 4-4-\frac{8k}{21} \\ 3+1+\frac{2k}{21} \\ k-2-\frac{4k}{21} \end{pmatrix} = \begin{pmatrix} -\frac{8k}{21} \\ 4+\frac{2k}{21} \\ -2+\frac{17k}{21} \end{pmatrix}$$

Hence, N is the point $\left(-\frac{8}{21}k, 4+\frac{2}{21}k, \frac{17}{21}k-2\right)$

(iii)

Direction vector of l_2 is $\begin{pmatrix} 0 \\ q \\ 2 \end{pmatrix}$

$$\cos 60^o = \frac{\left|\begin{pmatrix} 0 \\ q \\ 2 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}\right|}{\sqrt{q^2+4}\sqrt{5}}$$

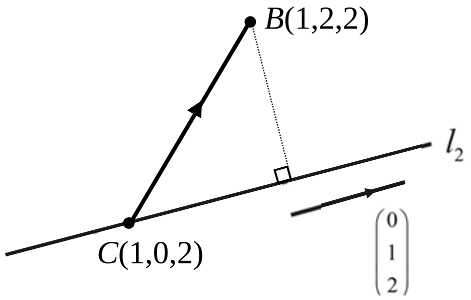
$$\frac{1}{2} = \frac{4}{\sqrt{5q^2+20}}$$

$$\sqrt{5q^2+20}=8$$

$$q^2=\frac{44}{5}$$

$q=\pm\sqrt{\frac{44}{5}}=\pm 2.97$ **(3 s.f.)**

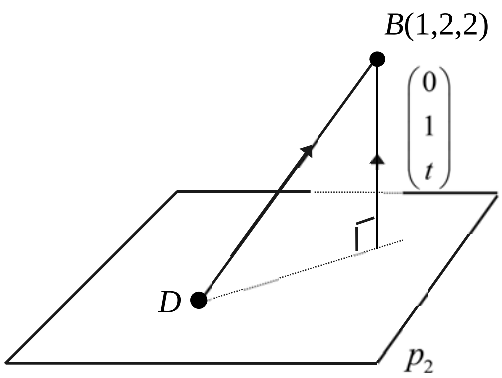
(iv)



$\overrightarrow{CB}=\begin{pmatrix}0\\2\\0\end{pmatrix}$

Shortest distance from B to the line l_2:

$$=\frac{\left|\begin{pmatrix}0\\2\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\2\end{pmatrix}\right|}{\sqrt{1+2^2}}=\frac{\left|\begin{pmatrix}4\\0\\0\end{pmatrix}\right|}{\sqrt{1+2^2}}=\frac{4}{\sqrt{5}}$$



Let D be any point on plane p_2.

$\overrightarrow{OD}=\begin{pmatrix}0\\0\\2\end{pmatrix}$ $\qquad\overrightarrow{DB}=\overrightarrow{OB}-\overrightarrow{OD}=\begin{pmatrix}1\\2\\0\end{pmatrix}$

	Shortest distance from B to plane p_2 $= \dfrac{\left\| \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ t \end{pmatrix} \right\|}{\sqrt{1+t^2}} = \dfrac{2}{\sqrt{1+t^2}}$ Since point $B(1,2,2)$ is equidistant from l_2 and p_2, $\dfrac{4}{\sqrt{5}} = \dfrac{2}{\sqrt{1+t^2}}$ $\sqrt{1+t^2} = \dfrac{\sqrt{5}}{2}$ $t^2 = \dfrac{1}{4}$ $t = \pm\dfrac{1}{2}$
10	(a) $\dfrac{\mathrm{d}v}{\mathrm{d}t} = 7.2 - kv^2$ When $v = 30, \dfrac{\mathrm{d}v}{\mathrm{d}t} = 5.4$ $5.4 = 7.2 - k\,(30^2)$ $k = 0.002$ $\dfrac{\mathrm{d}v}{\mathrm{d}t} = 7.2 - 0.002v^2$ $\int \dfrac{1}{7.2 - 0.002\,v^2}\,\mathrm{d}v = \int 1\,\mathrm{d}t$ $\dfrac{1}{0.002}\int \dfrac{1}{3600 - v^2}\,\mathrm{d}v = \int 1\,\mathrm{d}t$ $\dfrac{500}{2(60)}\ln\left\|\dfrac{60+v}{60-v}\right\| = t + C$ $\dfrac{25}{6}\ln\left\|\dfrac{60+v}{60-v}\right\| = t + C$ $\ln\left\|\dfrac{60+v}{60-v}\right\| = \dfrac{6}{25}t + \dfrac{6}{25}C$ $\dfrac{60+v}{60-v} = \pm e^{\frac{6}{25}t + \frac{6}{25}C}$ $\dfrac{60+v}{60-v} = Ae^{0.24t}, \ A = \pm e^{0.24C}$

	When $t=0, v=0$ $\therefore A=1$ $\therefore \dfrac{60+v}{60-v}=e^{0.24t}$ $60+v=(60-v)\,e^{0.24t}$ $60+v=60\,e^{0.24t}-ve^{0.24t}$ $v(1+e^{0.24t})=60\,e^{0.24t}-60$ $v=\dfrac{60\,(e^{0.24t}-1)}{(e^{0.24t}+1)}$ (b) $3x\dfrac{\mathrm{d}y}{\mathrm{d}x}=3y-4+x$ $y=ux$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=x\dfrac{\mathrm{d}u}{\mathrm{d}x}+u$ Sub eq (2) into eq (1) $3x(x\dfrac{\mathrm{du}}{\mathrm{d}x}+u)=3ux-4+x$ $3x^2\dfrac{\mathrm{d}u}{\mathrm{d}x}=x-4$ (shown) $\dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{x-4}{3x^2}$ $u=\int\left(\dfrac{1}{3x}-\dfrac{4}{3x^2}\right)\mathrm{d}x$ $=\dfrac{1}{3}\ln\lvert x\rvert+\dfrac{4}{3x}+C$ $\dfrac{y}{x}=\dfrac{1}{3}\ln\,x+\dfrac{4}{3x}+C$ $y=\dfrac{1}{3}x\ln\,x+\dfrac{4}{3}+Cx$ Reason: (1) According to the model, for positive values of C, the velocity would become very large/increase infinitely/increase to infinity . (2) For certain negative values of the arbitrary constant, the velocity would decrease to negative infinity.

11	$\pi r^2 h + \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = 300$ $\pi r^2\left(h + \frac{2}{3}r\right) = 300$ $h + \frac{2}{3}r = \frac{300}{\pi r^2}$ $h = \frac{300}{\pi r^2} - \frac{2}{3}r$ $C = \left(\pi r^2 + 2\pi rh\right)\left(\frac{a}{2}\right) + \frac{1}{2}\left(4\pi r^2\right)a$ $= \frac{5}{2}a\pi r^2 + a\pi rh$ $= \frac{5}{2}a\pi r^2 + a\pi r\left(\frac{300}{\pi r^2} - \frac{2}{3}r\right)$ $= \frac{5}{2}a\pi r^2 + \frac{300a}{r} - \frac{2a\pi r^2}{3}$ $= \frac{11}{6}a\pi r^2 + \frac{300a}{r}$ $\frac{\mathrm{d}C}{\mathrm{d}r} = \frac{11}{3}a\pi r - \frac{300a}{r^2}$ For minimum cost, $\frac{\mathrm{d}C}{\mathrm{d}r} = 0$ $\frac{11}{3}a\pi r - \frac{300a}{r^2} = 0$ $\frac{11}{3}a\pi r^3 = 300a$ $r^3 = \frac{900}{11\pi}$ $r = \sqrt[3]{\frac{900}{11\pi}}$

$$h = \frac{300}{\pi\left(\sqrt[3]{\frac{900}{11\pi}}\right)^2} - \frac{2}{3}\left(\sqrt[3]{\frac{900}{11\pi}}\right)$$

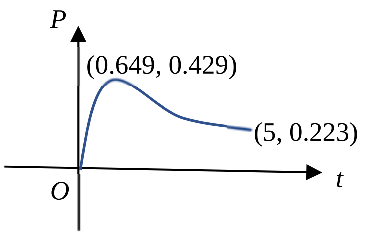
$$= \sqrt[3]{\frac{900}{11\pi}}\left[\frac{300}{\pi}\left(\frac{900}{11\pi}\right)^{-1} - \frac{2}{3}\right]$$

$$= \sqrt[3]{\frac{900}{11\pi}}\left[\frac{300}{\pi}\left(\frac{11\pi}{900}\right) - \frac{2}{3}\right]$$

$$= 3\sqrt[3]{\frac{900}{11\pi}}$$

$$\frac{\mathrm{d}^2C}{\mathrm{d}r^2} = \frac{11}{3}a\pi + \frac{600a}{r^3} > 0 \text{ as } a, r > 0$$

Thus, cost is a minimum when $r = \sqrt[3]{\frac{900}{11\pi}}$ and $h = 3\sqrt[3]{\frac{900}{11\pi}}$



$$\text{Area} = \int_0^5 \frac{\sqrt{\ln(t+1)}}{t+1}\,\mathrm{d}t$$

$$= \left[\frac{\left(\ln(t+1)\right)^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^5$$

$$= \frac{2}{3}[\ln 6]^{\frac{3}{2}}$$

The area represents the total profit made in the five years.